Career Compass AI: An AI-Powered Employability Assessment and Career Readiness Platform Using Resume Analysis and Skill Gap Assessment

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***Abstract*—**
Students often learn several technologies during their studies, but they may not know how closely their current skills match the requirements of a particular job. A resume can contain programming languages, frameworks, projects, internships, and certifications, but these details alone do not show whether the student has enough practical knowledge. Career Compass AI is proposed as an integrated platform that connects resume analysis, job matching, skill-gap identification, project recommendation, technical assessment, progress tracking, and career-roadmap generation.

The system accepts a student's resume together with a job description or selected target role. Important information is extracted and organized before the profile is compared with the selected requirements. The analysis then identifies matched skills, missing skills, and areas that need improvement. Important skill gaps are connected with practical project suggestions so that students receive a clearer direction for developing the required competencies.

The platform also provides role-specific assessments to obtain additional evidence of technical understanding. Previous assessment attempts are retained so that changes in performance can be observed over time. Based on the latest results, learning priorities and the career roadmap can be updated. In this way, Career Compass AI treats employability preparation as a continuous process of analysis, learning, assessment, and reassessment rather than as a one-time resume evaluation.

***Index Terms*—Career readiness, resume analysis, skill gap analysis, job matching, natural language processing, technical assessment, project recommendation, career roadmap**

I. Introduction

During their studies, students tend to gain knowledge of several technologies without understanding the correlation of their skills to the job requirements. A resume may include details such as programming languages, frameworks, projects, internships, and certificates, but this information alone does not guarantee practical knowledge. Career Compass AI is a comprehensive system offering services such as resume analysis, job matching, skills gap identification, project suggestions, technical assessments, progress tracking, and career roads planning.

The system accepts a student resume and the job description of a target position. Then, relevant information is extracted from the resume and organized before the resume is compared with the selected job requirements. The analysis identifies the skills matching job requirements, lacking skills, and the skills requiring improvement. Along with necessary skills gap information, students get appropriate project recommendations.

The platform provides assessments based on the specific role to ensure more evidence of technical knowledge. The history of assessments is preserved so that progress can be monitored. Following new results, learning priorities and the career pathway can be changed. Thus, Career Compass AI treats employability preparation as a perpetual process of analysis and learning rather than a singular examination of one's resume.

Preparing for campus placements can be challenging because students need to do more than just create a resume. They have to understand the required skills for a job, pinpoint their shortcomings in their profile, and choose what should be learned prior to applying for the position. This process becomes even more complicated when students look for options among various job titles because different positions might require different sets of programming languages, frameworks, databases, cloud technologies, and methodologies.

A resume contains relevant information about education and qualifications in terms of skills, projects worked on, internships, and certificates. However, listing a technology does not mean having enough experience to work with it. The wording used in a job description may differ from similar requirements. For instance, one company may use the term "RESTful services" while another one will refer to "REST API". An algorithm based only on exact wording may miss the necessary correlations between different skills.

Techniques provided by Natural Language Processing play a significant role in the processing this form of unstructured information. The implementation of Transformer-based frame-

works led to the introduction of contextual representations that could help resolve many issues facing people that require the processing of information in the field of natural language processing (NLP) [3]. BERT demonstrated the power of bidirectional into contextual representations for the improvement of understanding language [4]. Sentence-BERT made it possible to compare the meaning of sentences and fragments of texts in the least complicated manner, with the help of meaningful sentence embeddings [5]. The use of information retrieval technologies allows to represent documents and calculate their value [1].

The system links profile analysis with competence development. It matches the learner's profile with the selected goal position, pinpointing important qualifications gaps, proposing relevant projects to fill them and assessing the knowledge gained through specialized evaluations. Past results are saved for tracking progress and modifying future priorities.

II. Related Work

Automated recruitment as well as career services usually employ information retrieval, natural language processing, semantic similarity, machine learning as well as recommendation techniques. Information retrieval has provided the well-known techniques to describe documents and evaluate their relevancy to queries or documents related [1]. The learning to rank approaches also provide a good example of how to learn relevancy decisions.

It might also be true that transformers advanced the real capacity of NLP systems concerning the contextual relevance of words. BERT optimized this concept by allowing the use of bidirectional pre-training and getting impressive results in several cases of language comprehension. The Sentence-BERT method emerged later and allowed for comparing sentences using semantic embeddings. The rationale behind these methods is useful for employability application, mostly because the skills and responsibilities of a given position might be expressed by using different wording.

Random Forest has an ensemble learning method aimed at classifications and predictions [6]. XGBoost allows for creating an effective gradient boosting model for structured data [7]. This technique may be considered for willingness assessment when a labelled dataset is present.

Generally, existing methods relate only to different levels of career preparation process. For instance, a resume parsing software does not instruct the user on how to enhance their profile; a job matching system offers merely a score of similarity, while the way to proceed after that remains unclear; an evaluation tool assesses technical knowledge without linking it to the targeted job of a student.

In contrast, Career Compass AI integrates these steps into one workflow that allows conducting resume analysis, job matching, skill gap identification, project implementation, technical assessment, tracking progress, and adapting a roadmap.

III. Research Gap

The primary issue that this paper addresses is the gap in linking the resume-job match analysis with actual skill development.

Profile Analysis → Skill Diagnosis → Skill Development → Evaluation → Reevaluation

With this system the preparation stage of the student can be updated as new project or assessment evidence appears.

IV. Proposed System

Career Compass AI operates as a modular system with the result obtained from one step being used as an input for another step. The process starts from a student's CV and a job description or position of interest.

Currently available features are similar to a preparatory system rather than a one-time scoring system. Upon completion of the initial stage of analysis the produced outcomes are utilized to provide practical recommendations. Assessment results enter the process later on and are used to update the recommendations.

The main stages of Career Compass AI consist of resume analysis, job description analysis, normalization of skills, resume-job matching, identification of skill gaps, suggestion of projects, suggestion of a position, technical assessment, progress tracking, and building of a roadmap.

In the first phase, the resume of a student is converted into information that can be used by the remaining modules.

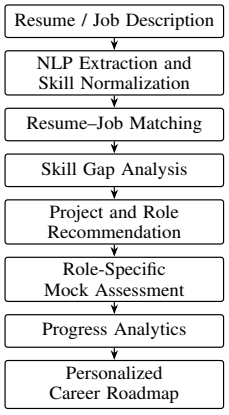


Fig. 1. Processing workflow of the proposed Career Compass AI framework.

V. Methodology

A. Resume Processing

The first stage converts the student's resume into information that can be used by the remaining modules. Resumes normally contain both structured headings and free-form descriptions. The system therefore extracts text and identifies sections such as education, technical skills, projects, internships, certifications, and achievements.

For machine-readable documents, text can be extracted directly. When scanned pages are present, OCR can be applied before further processing. The extracted text is cleaned by removing unnecessary spaces, repeated formatting characters, and other artifacts.

The resulting profile contains role-relevant attributes such as programming languages, tools, frameworks, databases, project experience, certifications, and internship information.

B. Job Description Processing

The job-description module follows a similar process. It identifies programming languages, frameworks, databases, cloud services, tools, and other competencies mentioned in the selected requirement.

Where sufficient information is available, requirements can be separated into mandatory and preferred categories. This helps the system prioritize important requirements over optional ones.

C. Skill Normalization

Skill normalization is important because the same competency may appear in several forms. Without normalization, a system may incorrectly identify a skill as missing even when an equivalent expression is present.

For example:

$$REST\ API,\ REST ful\ API,\ REST\ Services \rightarrow REST\ API \tag{1}$$

Similarly:

$$Amazon\ Web\ Services,\ AWS \rightarrow AWS \tag{2}$$

A normalization dictionary can map abbreviations, spelling variations, and common alternative expressions to a standard skill identifier.

D. Resume–Job Matching

Let S_r represent the normalized skills obtained from the resume and S_j represent the requirements obtained from the job description. A basic skill coverage value can be calculated as:

$$C = \frac{|S_r \cap S_j|}{|S_j|} \times 100 \tag{3}$$

where C represents the percentage of identified job requirements covered by the profile.

Keyword overlap may not be sufficient when different wording is used. Semantic representations can therefore be used in addition to direct matching. Cosine similarity between two embeddings is represented as:

$$Sim(E_r, E_j) = \frac{E_r \cdot E_j}{\|E_r\|\|E_j\|} \tag{4}$$

Here, E_r and E_j represent the corresponding vector representations.

VI. Skill Gap Analysis

After comparing the resume with the target role, the system identifies competencies that require attention. Instead of treating every unmatched term equally, the proposed approach divides the results into matched, missing, and improvement categories.

A matched skill is a competency represented in the student's profile. A missing skill is a requirement for which sufficient evidence is not found. An improvement skill is already present but may require stronger practical evidence or deeper understanding.

The priority of a gap depends on its importance to the selected role and the size of the observed gap. A conceptual priority score is:

$$P_s = R_s \times G_s \times I_s \tag{5}$$

where R_s represents role importance, G_s represents the estimated gap, and I_s represents relevance to the target position.

TABLE I
ILLUSTRATIVE SKILL GAP ANALYSIS

Skill	Status	Priority
Java	Match	–
Spring Boot	Match	–
MySQL	Match	–
Docker	Missing	High
AWS	Missing	High
Spring Security	Improve	Medium
Git	Improve	Medium

Note: The values shown are illustrative examples and are not experimental results.

VII. Project Recommendation

A conventional skill-gap report can identify what a student is missing but may not explain how that competency can be developed. Career Compass AI addresses this limitation by connecting identified gaps with practical project templates.

For example, when Docker, AWS, Spring Boot, and REST API development are identified as important gaps for a backend-oriented role, the platform can recommend an Employee Management System. The project can include a Spring Boot backend, MySQL database, REST endpoints, Docker containerization, Git-based version control, and cloud deployment.

The recommendation process can therefore be represented as:

Missing Skill → Project Template → Implementation Task → Skill Evidence

The purpose is not simply to increase the number of projects in a resume. The recommended project should provide an opportunity to apply the missing technology and create practical evidence of learning.

TABLE II
EXAMPLE PROJECT-TO-SKILL MAPPING

Project Component	Skill Developed
Backend Development	Spring Boot
Database Layer	MySQL
API Layer	REST API
Security	Spring Security
Containerization	Docker
Cloud Deployment	AWS
Version Control	Git

VIII. Suitable Job Role Recommendation

Students do not always have a single target position. A profile that is suitable for one role may require additional preparation for another. Career Compass AI can therefore compare the same profile with multiple predefined job roles.

The role recommendation output is intended to indicate preparation alignment rather than predict whether a company will hire a student. The comparison considers matched skills, missing requirements, and, where available, assessment performance.

TABLE III
ILLUSTRATIVE JOB ROLE SUITABILITY

Job Role	Match	Level
Backend Developer	92%	Strong
Java Developer	88%	Strong
Full Stack Developer	81%	Good
Data Analyst	68%	Moderate
AI Engineer	60%	Improve

Note: These values are illustrative and should not be reported as measured experimental performance.

IX. Role-Specific Mock Assessment

A resume can indicate that a student has used a particular technology, but it cannot independently verify the level of understanding. The proposed assessment module provides an additional source of evidence through role-specific questions.

The assessment is selected according to the target role and important skills identified during the earlier analysis. For a Backend Developer profile, the question set may include Java, Spring Boot, SQL, REST APIs, Docker, Git, and cloud fundamentals.

The assessment process identifies the target role, obtains high-priority skills, selects suitable questions, conducts the test, calculates overall and skill-wise scores, and stores the attempt for later comparison.

This distinction between a skill being mentioned and a skill being demonstrated is an important part of the proposed framework.

TABLE IV
ILLUSTRATIVE SKILL-WISE ASSESSMENT HISTORY

Skill	Test 1	Test 2	Change
Java	65%	88%	+23
Spring Boot	40%	75%	+35
SQL	78%	84%	+6
Docker	20%	70%	+50
AWS	30%	65%	+35

X. Skill Analytics and Progress Tracking

Career preparation becomes more meaningful when changes can be observed over time. The proposed system stores individual assessment attempts instead of retaining only the latest score.

For a particular skill s, the change between two attempts can be represented as:

$$\Delta_s = S_{latest,s} - S_{previous,s} \tag{6}$$

where $S_{latest,s}$ is the latest score and $S_{previous,s}$ is an earlier score.

For example, an increase in the Docker assessment score from 20% to 70% corresponds to a 50 percentage-point improvement in assessment performance. This should not be interpreted as a direct measurement of real-world proficiency.

The analytics component can display the initial score, latest score, change over time, strongest skill, weakest skill, and remaining high-priority gaps. This gives students a simple way to decide whether their current preparation strategy needs to continue or change.

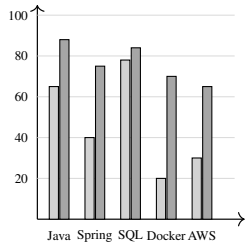


Fig. 2. Illustrative comparison of skill scores across assessment attempts.

XI. Personalized and Adaptive Career Roadmap

The roadmap is the stage where the analysis becomes a practical preparation plan. Instead of presenting students with a long list of unrelated learning resources, the system organizes activities around skills that are important for the selected role.

A typical roadmap begins with understanding the fundamentals of an identified skill, followed by suitable learning material and a practical task. The student can then apply the skill in a project, complete a role-specific assessment, review the result, and reassess the remaining gaps.

For example, if a student has reasonable Java knowledge but performs poorly in Docker and AWS, the next roadmap stage can focus on Docker fundamentals, containerizing a Spring Boot application, understanding basic AWS deployment concepts, and repeating the assessment.

The roadmap therefore changes according to newly observed evidence rather than remaining fixed after the initial resume analysis.

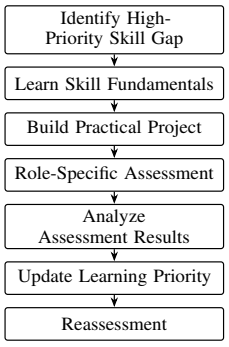


Fig. 3. Adaptive career-roadmap sequence driven by skill-gap and assessment feedback.

XII. Preparation Readiness Model

The system can provide a preparation-readiness indicator by combining observations from the student's profile, target role, and assessment performance. Unlike a simple resume

similarity value, the indicator considers multiple sources of evidence.

A conceptual formulation is:

$$R = w_1 M + w_2 C + w_3 A \tag{7}$$

where M represents resume-job matching, C represents important skill coverage, A represents assessment performance, and w_1, w_2, w_3 are configurable weights.

The weights satisfy:

$$w_1 + w_2 + w_3 = 1 \tag{8}$$

The indicator is intended to describe preparation status rather than predict employment. Before being treated as a calibrated measure, the formulation should be tested with a suitable dataset.

Recruitment decisions depend on several factors outside the scope of this model, including interviews, coding performance, communication ability, work experience, and organization-specific selection procedures.

XIII. System Architecture and Implementation

The proposed platform follows a modular web architecture. The user interface provides facilities for resume upload, target-role selection, analysis results, assessment interaction, progress visualization, and roadmap presentation.

The application layer coordinates communication between the interface and processing modules. NLP processing handles resume and job-description text, while the matching engine compares normalized skills and semantic representations. The skill-gap module identifies missing and improvement areas. The recommendation engine connects these gaps with suitable projects, while the assessment engine manages role-specific tests.

A persistent database retains user profiles, extracted skills, target roles, assessment attempts, recommendations, and roadmap information. This structure allows previous results to be used when generating later recommendations.

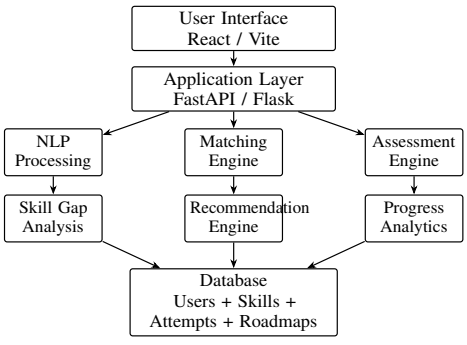


Fig. 4. Modular architecture of the proposed Career Compass AI platform.

A. Technology Stack

The implementation can use React with Vite for the web interface and a Python-based backend such as FastAPI or Flask for application services. Python libraries can support document processing, NLP operations, semantic comparison, and machine-learning components.

A relational database such as MySQL or PostgreSQL can store user profiles, extracted skills, job requirements, assessment attempts, project recommendations, and roadmap information. Chart.js or Recharts can be used to display skill-progress information through simple visualizations.

TABLE V
PROPOSED TECHNOLOGY STACK

Layer	Technology
Frontend	React.js, Vite
UI Styling	CSS / Tailwind CSS
Backend	FastAPI / Flask
NLP	Python, NLP Libraries
ML	Scikit-learn
Database	MySQL / PostgreSQL
Visualization	Chart.js / Recharts
Version Control	Git / GitHub

XIV. Data Model and Processing Flow

The database is designed to preserve information required for repeated analysis. A student may upload more than one resume and evaluate the profile against different target roles. Each target role can contain several required skills, and each assessment can generate multiple skill-wise results.

The principal entities include the user, resume, job profile, skill, skill gap, assessment, attempt, recommendation, and roadmap. The user entity stores basic account information, while the resume entity stores uploaded resume information and extracted content. The job profile contains the selected role and job-description information.

Skill-gap records maintain matched, missing, and improvement classifications. Assessment and attempt records preserve test information and individual results. Recommendation and roadmap records store suggested projects and learning priorities.

Maintaining these relationships allows the system to use earlier information when generating later recommendations. Assessment history, in particular, provides evidence for determining whether a skill area is improving or continues to require attention.

XV. Evaluation Methodology

Evaluation of Career Compass AI should be performed at both module and end-to-end levels. Since the platform performs several different tasks, a single accuracy value would not adequately describe its overall behaviour.

A. Resume Extraction Evaluation

Resume extraction can be evaluated using manually annotated resumes. Precision, recall, and F1-score can be calculated for information such as technical skills, education, experience, and certifications.

$$Precision = \frac{TP}{TP + FP} \tag{9}$$

$$Recall = \frac{TP}{TP + FN} \tag{10}$$

$$F1 = 2\frac{Precision \times Recall}{Precision + Recall} \tag{11}$$

These measures indicate how closely automatically extracted information agrees with manually verified information.

B. Skill Matching Evaluation

For skill matching, resume-job pairs can be manually annotated. Each requirement can be labelled according to whether the student's profile satisfies it. System predictions can then be compared with these reference labels.

C. Recommendation Evaluation

Project recommendations should be evaluated according to relevance. Evaluators can examine whether a recommended project addresses the identified skill gap and provides a realistic opportunity to apply the missing competency.

D. Assessment Evaluation

The assessment module can be evaluated in terms of question relevance, skill coverage, difficulty consistency, and repeated-attempt behaviour. Questions should remain aligned with the competencies expected for the selected role.

TABLE VI
EVALUATION PARAMETERS AND THEIR PURPOSE

Parameter	Evaluation Purpose
Resume Extraction	Correct extraction of profile entities
Skill Identification	Correct detection of relevant skills
Job Matching	Alignment with job requirements
Gap Detection	Identification of missing competencies
Recommendation	Relevance of suggested projects
Assessment	Reliability across attempts
Roadmap	Alignment with identified weaknesses
Progress Tracking	Measurement of score changes

XVI. Baseline and Ablation Study

An ablation study can be used to understand the contribution of the major modules. A simple baseline can begin with resume-job matching alone. Additional components can then be introduced progressively.

The first configuration performs profile matching only. The second adds skill-gap identification. The third includes role-specific assessment, while the final configuration adds the adaptive roadmap. The purpose is to determine whether the additional modules provide useful information beyond a conventional resume-matching approach.

TABLE VII
ABLATION CONFIGURATIONS

ID	Match	Gap	Test	Roadmap	Purpose
A	Yes	No	No	No	Baseline
B	Yes	Yes	No	No	Gap
C	Yes	Yes	Yes	No	Assessment
D	Yes	Yes	Yes	Yes	Complete

XVII. Results and Analysis Plan

The numerical performance of the proposed system should be reported only after testing with an appropriate dataset. The illustrative values shown in this paper demonstrate the intended presentation and should not be interpreted as measured experimental results.

The planned evaluation can include resume-extraction precision, recall and F1-score, skill-identification accuracy, agreement between automated job matching and human annotation, relevance of identified skill gaps, usefulness of project recommendations, assessment consistency, and changes in skill-wise scores across repeated attempts.

An important experiment is to compare the roadmap before and after a student completes an assessment. If the assessment shows improvement in one skill while another remains weak, the next roadmap should reflect that change. This provides a direct way to examine whether the adaptive component responds to newly available evidence.

XVIII. Discussion

The main strength of Career Compass AI is that it connects several stages of career preparation rather than treating resume analysis as the final result. A matching score can indicate how closely a profile resembles a job requirement, but it does not explain how the student can improve the profile.

The proposed workflow addresses this issue by converting identified gaps into practical development activities. When an important skill is missing, the system can prioritize it and connect it with a suitable project. The student can then use the project to gain implementation experience before taking another assessment.

The assessment component adds another useful distinction. A technology appearing in a resume is not necessarily equivalent to demonstrated knowledge. By evaluating role-specific questions and retaining previous attempts, the system obtains an additional signal that can be used when generating the roadmap.

The framework can also support multiple career directions. A student may have stronger preparation alignment with a Backend Developer role than an AI Engineer role. Instead of forcing a single recommendation, the system can show relative preparation levels and identify skills required to move toward another role.

The proposed framework should therefore be viewed as a career-preparation support system rather than an automated recruitment decision system.

XIX. Limitations

The proposed framework has several limitations. The first is the quality of input documents. Poorly formatted resumes, scanned pages, unusual layouts, or ambiguous terminology can reduce the accuracy of text and skill extraction.

A second limitation is that semantic similarity cannot prove practical proficiency. A student may mention a technology in a resume without having sufficient hands-on experience. The assessment module provides an additional signal, but even a structured test cannot completely replace practical project evaluation.

Recommendation quality also depends on the underlying skill-to-project knowledge base. If a new technology or specialized role is not adequately represented, the platform may produce a less useful recommendation.

The preparation-readiness indicator should not be interpreted as a probability of employment. Actual recruitment depends on interviews, coding ability, communication, experience, company-specific expectations, and other factors outside the proposed model.

Reliable evaluation also requires a sufficiently diverse dataset containing resumes, job descriptions, target roles, and manually verified skill relationships. Broader validation is therefore required before making claims about generalization across institutions or industries.

XX. Future Work

Several extensions can improve Career Compass AI. Large language models can be investigated for more detailed contextual extraction of skills and responsibilities from resumes and job descriptions. They may also provide explanations for why a particular skill has been prioritized.

A skill knowledge graph is another possible extension. Such a graph could represent relationships among roles, technologies, prerequisite skills, projects, and learning activities. This would allow the system to recommend a suitable sequence of skills rather than treating every missing competency independently.

The assessment module can also be made adaptive. When a student consistently performs well on beginner-level questions, the system can gradually introduce questions of greater difficulty. Practical coding exercises can provide another source of evidence beyond multiple-choice questions.

Future versions may also consider project portfolios, Git repositories, internship experience, interview preparation, and communication-related skills. At the institutional level, anonymized aggregate skill-gap information could help placement departments identify commonly missing competencies and design targeted training programmes.

XXI. Conclusion

Career Compass AI presents an integrated approach to student employability preparation by combining resume analysis, job-description matching, skill-gap identification, practical project recommendation, role-specific assessment, progress analytics, and personalized roadmap generation.

The central idea is to move beyond a one-time resume score. The student's current profile is first compared with a selected target role. The system then identifies important missing or weak competencies and connects them with practical development activities. After the student completes relevant preparation, the assessment module provides another source of information about current knowledge.

Retaining previous assessment attempts allows the system to observe changes rather than relying only on the latest result. These observations can be used to modify learning priorities and produce a more relevant roadmap for the next preparation stage.

The framework is therefore intended to support an iterative process of analysis, learning, assessment, and reassessment. The proposed methodology provides a foundation for a student-centred employability platform, while future experiments with diverse and manually validated datasets are required to quantify the effectiveness of each component.

XXII. *

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